or 1 cm represents 50,000 cm or 0.5 Km (1 Km = 100000 cm). Therefore, 10 cm represents 5 km

Construction

The graphical scale may be constructed by following these steps:

Draw a straight line of 10 cm and divide it into 5 equal parts these are the primary division. Mark the first division as 0. Assign the value of 1 km for the four divisions starting from 0. Therefore the primary scale has 4 divisions and is 4 km long.

Divide the extreme left side division into 10 equal parts and mark each division by a value of 100 metres, beginning from 0. This is the secondary scale representing 1000 mts.



Exercise

1. Convert the statement into RF.
   a. 1 cm to 10 km
   b. 1 cm to 5 km
   c. 1 cm to 1 km
   d. 1 cm to 50km
   e. 1 cm to 100 km
2. Convert the RF into statement:
   a. 1: 100000
   b. 1: 50000
   c. 1: 250000
   d. 1: 5000000
   e. 1: 30000
3. Construct a graphical scale for the following:
   a. 1 cm to 10 km
   b. 1 cm to 5 km
   c. 1 cm to 1 km
   d. 1 cm to 50km
   e. 1 cm to 100 km

9.3. Measurement of distance between places

The linear features shown on the maps can be classified into two broad categories, i.e.

1.Straight lines

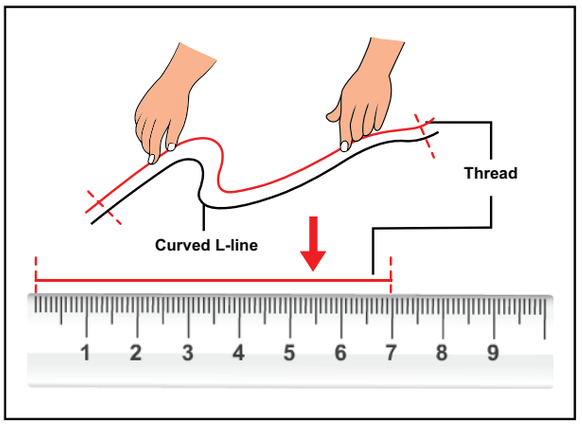
2. Erratic or zigzag lines.

Straight line features

The measurement of straight line features like roads, railway lines and canals is simple. It can be taken directly with a pair of dividers or a scale placed on the map surface.

Erratic or zig zag lines

However, care should be taken to measure distances , along erratic paths, such as roads, boundaries, coastlines, rivers and streams. The distances along all such features can be measured by placing a thread at the starting point and carrying it along the line up to the end point. The thread is then stretched and measured to determine the distance. It can also be measured by using a simple instrument called Rotameter. The wheel of the 'rotameter' is moved along the route or line from start to end and the reading noted down.



Measuring a curved line using a thread and a ruler

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